

17.02.2024

Module-2

Numerical methods to solve the system of Linear Equations.

* Matrix partition method:

Partition method is used to find the inverse of a matrix. Let the coefficient matrix A be partitioned as

$$A = \begin{bmatrix} B & C \\ E & D \end{bmatrix}_{n \times n}$$

Where B is a $r \times r$ matrix

C is a $r \times s$ matrix

E is a $s \times r$ matrix

D is a $s \times s$ matrix.

Here, r and s are +ve integers with $r+s=n$. Further, we also assume that A^{-1} is partitioned similarly as follows

$$A^{-1} = \begin{bmatrix} X & Y \\ Z & V \end{bmatrix}_{n \times n}$$

Where X, Y, V, Z are the matrices of the same order as of B, C, D, E .

$$\text{Now, we have } AA^{-1} = \begin{bmatrix} B & C \\ E & D \end{bmatrix} \begin{bmatrix} X & Y \\ Z & V \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & I_s \end{bmatrix}$$

$$\Rightarrow BX + CZ = I_r \quad \text{--- (1)} \quad \rightarrow (*)$$

$$BY + CV = 0 \text{ (order - } r \times s) \quad \text{--- (2)}$$

$$EX + DZ = 0 \text{ (order - } s \times r) \quad \text{--- (3)}$$

$$EY + DV = I_s \quad \text{--- (4)}$$

$$(2) \Rightarrow BY + CV = 0$$

$$Y = -B^{-1}CV \quad \text{--- (5)}$$

Substituting (5) in (4):

$$(5) \Rightarrow EY + DV = I_s$$

$$E(-B^{-1}CV) + DV = I_s$$

$$-EB^1CV + DV = I_s$$

$$V(-EB^1C + D) = I_s$$

$$V = (D - EB^1C)^{-1} \quad \text{--- (6)}$$

$$(3) \Rightarrow EX + DZ = 0$$

$$X = -E^{-1}DZ \quad \text{--- (7)}$$

Substituting (7) in (1):

$$(1) \Rightarrow BX + CZ = I_r$$

$$B(-E^{-1}DZ) + CZ = I_r$$

$$Z(-BE^{-1}D + C) = I_r$$

$$Z = (C - BE^{-1}D)^{-1}$$

It is obvious that we need to find two inverse matrices i.e., B^{-1} and $(D - EB^1C)^{-1}$ of orders r and s respectively.

$$\Rightarrow Z = -VEB^{-1}$$

1. Use the matrix partition method to find the solution of system of equations

$$x_1 + x_2 + x_3 = 1$$

$$4x_1 + 3x_2 - x_3 = 6$$

$$3x_1 + 5x_2 + 3x_3 = 4$$

$$A = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array} \right]_{3 \times 3}$$

Let the matrix A be partitioned as

$$A = \left[\begin{array}{cc|c} B & C \\ E & D \end{array} \right] \Rightarrow A^{-1} = \left[\begin{array}{cc} X & Y \\ Z & V \end{array} \right]$$

$$B = \left[\begin{array}{cc} 1 & 1 \\ 4 & 3 \end{array} \right] \Rightarrow B^{-1} = \frac{1}{|B|} \left[\begin{array}{cc} 3 & -1 \\ -4 & 1 \end{array} \right]$$

$$B^{-1} = - \left[\begin{array}{cc} 3 & -1 \\ -4 & 1 \end{array} \right]$$

$$V = (D - EB^{-1}C)^{-1}$$

$$= 3 + [3 \ 5] \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= 3 + [-11 \ 2] \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= (3 - 13)^{-1}$$

$$= (-10^{-1})$$

$$V = \frac{-1}{10}$$

$$Y = -B^{-1}CY$$

$$= \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} -1 \\ 10 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ -5 \end{bmatrix} (-1/10)$$

$$Y = \begin{bmatrix} -0.4 \\ 0.5 \end{bmatrix}$$

$$Z = \cancel{VD}B^{-1} - VE B^{-1}$$

$$= \frac{-1}{10} [3 \ 5] \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix}$$

$$= \frac{-1}{10} [-11 \ 2]$$

$$Z = [1.1 \ -0.2]$$

$$X = B^{-1} - B^{-1}CZ$$

$$= \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix} + \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} [1.1 \ -0.2]$$

$$= \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix} + \begin{bmatrix} 4.4 & -0.8 \\ -5.5 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1.4 & 0.2 \\ -1.5 & 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} x & y \\ z & v \end{bmatrix} = \begin{bmatrix} 1.4 & 0.2 & -0.4 \\ -1.5 & 0 & 0.5 \\ 1.1 & -0.2 & -0.1 \end{bmatrix}$$

~~$$AX = B$$~~
~~$$X = A^{-1}B$$~~

$$AX = B$$

$$X = A^{-1}B = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \& \quad B = \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$$

$$X = \begin{bmatrix} 1.4 & 0.2 & -0.4 \\ -1.5 & 0 & 0.5 \\ 1.1 & -0.2 & -0.1 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \\ -0.5 \end{bmatrix}$$

$$\therefore x_1 = 1, \quad x_2 = 0.5, \quad x_3 = -0.5.$$

2. $A = \begin{bmatrix} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{bmatrix}$

Find the inverse of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{bmatrix} \quad \text{using partition method \& hence solve the system of equations where}$$

$$B = \begin{bmatrix} -10 \\ 8 \\ 7 \\ -5 \end{bmatrix}$$

$$A = \left[\begin{array}{cc|cc} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{array} \right]$$

$$\text{then } B = \begin{bmatrix} 2 & 1 \\ 4 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 \\ 2 & -1 \end{bmatrix} \quad E = \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix}$$

$$B^{-1} = \frac{\text{adj } B}{|B|} = \frac{-1}{4} \begin{bmatrix} 0 & -1 \\ -4 & -2 \end{bmatrix}$$

$$V = (D - EB^{-1}C)^{-1}$$

$$V = \begin{bmatrix} 2 & 0 \\ 2 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1/4 \\ 1 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 2 & 0 \\ 2 & -1 \end{bmatrix} - \begin{bmatrix} 1.5 & -4.25 \\ 0.5 & -7.25 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.5 & 4.25 \\ 1.5 & 6.25 \end{bmatrix}^{-1}$$

$$V = \begin{bmatrix} -1.923 & 1.3076 \\ 0.4615 & -0.153 \end{bmatrix}$$

$$Y = -B^{-1}CV$$

$$Y = - \begin{bmatrix} 0 & 1/4 \\ 1 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1.923 & 1.3076 \\ 0.4615 & -0.153 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0.8461 & -0.615 \\ 1.1537 & -0.382 \end{bmatrix}$$

$$Z = -V^T B^{-1}$$

$$Z = - \begin{bmatrix} -1.923 & 1.3076 \\ 0.4615 & -0.153 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1/4 \\ 1 & -1/2 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1.3076 & -0.346 \\ -0.153 & -0.077 \end{bmatrix} \begin{bmatrix} -0.076 & 1.1537 \\ -0.464 & -0.075 \end{bmatrix}$$

$$X = B^{-1} - B^{-1}CZ$$

$$X = \begin{bmatrix} 0 & 1/4 \\ 1 & -1/2 \end{bmatrix} - \begin{bmatrix} 0 & 1/4 \\ 1 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1.3076 & -0.346 \\ -0.153 & -0.077 \end{bmatrix}$$

$$X = \begin{bmatrix} -0.154 & -0.308 \\ -0.16 & -0.687 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} X & Y \\ Z & V \end{bmatrix} = \begin{bmatrix} -0.154 & -0.308 & 0.8461 & -0.615 \\ -0.16 & -0.687 & 1.1537 & -0.382 \\ -0.076 & 1.1537 & -1.923 & 1.3076 \\ -0.464 & -0.075 & 0.4615 & -0.153 \end{bmatrix}$$

$$AX = B \Rightarrow X = A^{-1}B$$

$$X = \begin{bmatrix} 0.154 & -0.308 & 0.8461 & -0.615 \\ -0.16 & -0.687 & 1.1537 & -0.382 \\ -0.076 & 1.1537 & -1.923 & 1.3076 \\ -0.464 & -0.075 & 0.4615 & -0.153 \end{bmatrix} \begin{bmatrix} -10 \\ 8 \\ 7 \\ 5 \end{bmatrix}$$

$$X = \begin{bmatrix} -1.1563 \\ 2.2699 \\ 3.0666 \\ 6.5055 \end{bmatrix} \quad X = \begin{bmatrix} 4.9937 \\ 6.0899 \\ -10.0094 \\ 8.0355 \end{bmatrix}$$

$$\therefore x_1 = -1.1563, x_2 = 2.2699, x_3 = 3.0666, x_4 = 6.5055.$$

3. Find the inverse of the matrix using partition method, $A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix}, D = \begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}, E = \begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

$$V = (D - EB^{-1}C)^{-1}$$

$$V = \left(\begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix} \right)^{-1}$$

$$V = \begin{bmatrix} -1 & 1 \\ -2 & 3 \end{bmatrix}$$

$$Z = -VEB^{-1}$$

$$Z = \begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}$$

$$X = B^{-1} - B^{-1}CZ$$

$$X = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} X & Y \\ Z & V \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -3 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{bmatrix}$$

$$X = A^{-1}B$$

$$X = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -3 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$$

$$X = \begin{bmatrix} -2 \\ -10 \\ 4 \\ 11 \end{bmatrix}$$

$$\therefore x_1 = -2, x_2 = -10, x_3 = 4, x_4 = 11$$

* Cholesky method :

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

⋮

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_n$$

$$AX = B \quad \text{--- (1)}$$

$$A = LL^T \quad \text{--- (2)}$$

where $L = l_{ij}$, if $i < j = 0$

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

Sub (2) in (1) :

$$LL^T X = B \quad \text{--- (3)}$$

$$\text{Put } L^T X = Y \quad \text{--- (4)}$$

Sub (4) in (3) :

$$LY = B \quad \text{--- (5)}$$

Using (5) and solving by forward substitution,
we get y_i

Now by solving eqn (4) by backward
substitution we get required solution.

To find inverse :

$$A = LL^T$$

$$A^{-1} = (LL^T)^{-1}$$

$$A^{-1} = (L^T)^{-1} L^{-1}$$

$$A^{-1} = (L^{-1})^T L^{-1}$$

To find L^{-1} : $LL^{-1} = I$.

1. Solve the system of equations, using cholesky method :

$$x + 2y + 3z = 5$$

$$2x + 8y + 22z = 6$$

$$3x + 22y + 82z = -10$$

The given system of equations in matrix form

$$AX = B$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}$$

We consider $A = LL^T$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{12} & l_{13} \\ 0 & l_{22} & l_{23} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{11}l_{12} & l_{11}l_{13} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

$$l_{11}^2 = 1 \Rightarrow l_{11} = 1$$

$$l_{11}l_{21} = 2 \Rightarrow l_{21} = 2$$

$$l_{11}l_{31} = 3 \Rightarrow l_{31} = 3$$

$$l_{21}^2 + l_{22}^2 = 8 \Rightarrow l_{22} = 2$$

$$l_{21}l_{31} + l_{22}l_{32} = 22 \Rightarrow 6 + 2l_{32} = 22 \Rightarrow l_{32} = 8$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 82 \Rightarrow 9 + 64 + l_{33}^2 = 82 \Rightarrow l_{33} = 3$$

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix}$$

$$\text{Consider } AX = B \Rightarrow LL^T X = B$$

$$\text{Put } L^T X = Y$$

$$\Rightarrow LY = B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}$$

$$y_1 = 5$$

$$2y_1 + 2y_2 = 6$$

$$3y_1 + 8y_2 + 3y_3 = -10$$

Solving by forward substitution method :

$$2(5) + 2y_2 = 6$$

$$2y_2 = -4 \Rightarrow y_2 = -2$$

$$3(5) + 8(-2) + 3y_3 = -10$$

$$3y_3 = -9 \Rightarrow y_3 = -3$$

Consider $L^T X = Y$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$$

$$x + 2y + 3z = 5$$

$$2y + 8z = -2$$

$$3z = -3$$

$$\Rightarrow z = -1$$

$$\rightarrow 2y - 8 = -2$$

$$2y = 6 \Rightarrow y = 3$$

$$\rightarrow x + 6 - 3 = 5$$

$$x = 2$$

$\therefore$ The required solutions are $x=2, y=3, z=-1$.

To find A^{-1} :

$$A = LL^T$$

$$A^{-1} = (LL^T)^{-1} = (L^T)^{-1} L^{-1}$$

$$A^{-1} = (L^{-1})^T L^{-1}$$

$$\text{WKT, } LL^{-1} = I$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix} \begin{bmatrix} l_{11}^* & 0 & 0 \\ l_{21}^* & l_{22}^* & 0 \\ l_{31}^* & l_{32}^* & l_{33}^* \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} l_{11}^* & 0 & 0 \\ 2l_{11}^* + 2l_{21}^* & 2l_{22}^* & 0 \\ 3l_{11}^* + 8l_{21}^* + 3l_{31}^* & 0 + 8l_{22}^* + 3l_{32}^* & 3l_{33}^* \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$l_{11}^* = 1$$

$$2l_{11}^* + 2l_{21}^* = 0 \Rightarrow l_{21}^* = -1$$

$$2l_{22}^* = 1 \Rightarrow l_{22}^* = 1/2$$

$$3l_{11}^* + 8l_{21}^* + 3l_{31}^* = 0 \Rightarrow l_{31}^* = 5/3$$

$$8l_{22}^* + 3l_{32}^* = 0 \Rightarrow l_{32}^* = -4/3$$

$$3l_{33}^* = 1 \Rightarrow l_{33}^* = 1/3$$

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/2 & 0 \\ 5/3 & -4/3 & 1/3 \end{bmatrix}$$

$$A^{-1} = (L^{-1})^T L^{-1}$$

$$A^{-1} = \begin{bmatrix} 1 & -1 & 5/3 \\ 0 & 1/2 & -4/3 \\ 0 & 0 & 1/3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/2 & 0 \\ 5/3 & -4/3 & 1/3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 43/9 & -49/18 & 5/9 \\ -49/18 & 73/36 & -4/9 \\ 5/9 & -4/9 & 1/9 \end{bmatrix}$$

2. Solve the system of equations, using cholesky method and also find A^{-1} .

$$4x_1 - x_2 = 1$$

$$-x_1 + 4x_2 - x_3 = 0$$

$$-x_2 + 4x_3 - x_4 = 0$$

$$-x_3 + 4x_4 = 0$$

04.03.2025

Module - 4 (CHOICE)

Eigen Value & Eigen Vector in Symmetric Matrix

* Jacobi Method :

$$A = \begin{bmatrix} a_{ii} & a_{ik} \\ a_{ki} & a_{kk} \end{bmatrix}$$

Orthogonal matrix : $S = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

$$S^{-1}AS = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a_{ii} & a_{ik} \\ a_{ki} & a_{kk} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta a_{ii} + \sin \theta a_{ki} & \cos \theta a_{ik} + \sin \theta a_{kk} \\ -\sin \theta a_{ii} + \cos \theta a_{ki} & -\sin \theta a_{ik} + \cos \theta a_{kk} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \theta a_{ii} + \sin \theta \cos \theta a_{ki} + \cos \theta \sin \theta a_{ik} + \sin^2 \theta a_{kk} & -\cos \theta \sin \theta a_{ii} - \sin^2 \theta a_{ki} + \cos^2 \theta a_{ik} + \sin \theta \cos \theta a_{kk} \\ -\sin \theta \cos \theta a_{ii} + \cos^2 \theta a_{ki} + \sin^2 \theta a_{ik} + \cos \theta \sin \theta a_{kk} & \sin^2 \theta a_{ii} + \cos \theta \sin \theta a_{ki} - \sin \theta \cos \theta a_{ik} + \cos^2 \theta a_{kk} \end{bmatrix}$$

$$\begin{aligned} -\cos \theta \sin \theta a_{ii} - \sin^2 \theta a_{ki} + \cos^2 \theta a_{ik} + \sin \theta \cos \theta a_{kk} &= 0 \\ \sin \theta \cos \theta (a_{kk} - a_{ii}) + a_{ik} (-\sin^2 \theta + \cos^2 \theta) &= 0 \\ \frac{1}{2} \sin 2\theta (a_{kk} - a_{ii}) + a_{ik} (\cos 2\theta) &= 0 \end{aligned}$$

$$a_{ik} (\cos 2\theta) = -\frac{1}{2} \sin 2\theta (a_{kk} - a_{ii})$$

$$\frac{2a_{ik}}{a_{ii} - a_{kk}} = \frac{\sin 2\theta}{\cos 2\theta} = \tan 2\theta$$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{ik}}{a_{ii} - a_{kk}} \right)$$

Problems:

1. For the symmetric matrix $A = \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}$, Find the eigen value and vector.

$$A = \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix}$$

The largest non-diagonal element $a_{13} = 2$
 $a_{11} = 1$, $a_{33} = 1$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{ik}}{a_{ii} - a_{kk}} \right)$$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{4}{0} \right)$$

$$\theta = \frac{1}{2} \times \frac{\pi}{2} = \frac{\pi}{4}$$

$$S_1 = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$B_1 = S_1^T A S_1 = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}^T \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$B_1 = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Here, the largest non-diagonal element $a_{12} = 2$
 $a_{11} = 3$, $a_{22} = 3$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{ik}}{a_{ii} - a_{kk}} \right) \Rightarrow \theta = \frac{\pi}{4}$$

CLASSMATE
Date _____
Page _____

$$S_1 = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_2 = S_2^T B_1 S_1 = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 3 & \sqrt{2} \\ 2 & \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_2 = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{It is a diagonal matrix.}$$

$\therefore$ Eigen values are 5, 1, -1

The corresponding eigen vectors are

$$S = S_1 S_2 = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 0.5 & -0.5 & -0.7071 \\ 0.7071 & 0.7071 & 0 \\ 0.5 & -0.5 & 0.7071 \end{bmatrix}$$

2. Find the Eigen value and Eigen vector for the symmetric matrix using Jacobi method.

$$A = \begin{bmatrix} 2 & 3 & 1/\sqrt{2} \\ 3 & 2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 5 \end{bmatrix}$$

The largest non-diagonal element is $a_{12} = 3$
 $a_{11} = 2, \quad a_{22} = 2$

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{ik}}{a_{ii} - a_{kk}} \right)$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$S_1 = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_1 = S_1^{-1} A S_1 = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 3 & 1/\sqrt{2} \\ 3 & 2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 5 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B_1 = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 5 \end{bmatrix}$$

Here, the largest non-diagonal element $a_{13} = 1$
 $a_{11} = 5$, $a_{33} = 5$.

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2a_{ik}}{a_{ii} - a_{kk}} \right)$$

$$\theta = \frac{\pi}{4}$$

$$S_2 = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$B_2 = S_2^{-1} B_1 S_2 = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}^{-1} \begin{bmatrix} 5 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$B_2 = \begin{bmatrix} 6 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$\therefore$ Eigen values are 6, -1, 4.

The corresponding Eigen vectors are

$$S = S_1 S_2 = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}$$

$$S = \begin{bmatrix} 0.5 & -0.7071 & -0.5 \\ 0.5 & 0.7071 & -0.5 \\ 0.7071 & 0 & 0.7071 \end{bmatrix}$$

* Given's method:

'A' is a 3x3 symmetric matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$\begin{matrix} \text{key element} \\ \text{pivotal element} \end{matrix}$

$$a_{ik} : a_{ii} \ a_{ik} \ a_{ki} \ a_{kk}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\tan \theta = \frac{a_{13}}{a_{12}}$$

$$B = S_1^{-1} A S_1 = \begin{bmatrix} a_1 & b_2 & 0 \\ b_2 & a_2 & b_3 \\ 0 & b_3 & a_3 \end{bmatrix} \text{ is a trigonal matrix}$$

$$|B - \lambda I| = 0 \quad (\text{Characteristic equation})$$

$$\begin{vmatrix} a_1 - \lambda & b_2 & 0 \\ b_2 & a_2 - \lambda & b_3 \\ 0 & b_3 & a_3 - \lambda \end{vmatrix} = 0$$

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda$$

$$P_{k+1}(\lambda) = (a_{k+1} - \lambda) P_k(\lambda) - b_{k+1}^2 P_{k-1}(\lambda)$$

$$k = 1, 2, 3, \dots$$

$$(a_1 - \lambda)x_1 + b_2x_2 + 0x_3 = 0$$

$$b_2x_1 + (a_2 - \lambda)x_2 + b_3x_3 = 0$$

$$0x_1 + b_3x_2 + (a_3 - \lambda)x_3 = 0$$

1. Find the eigen value of the symmetric matrix using given's method.

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 3 & 1 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$\tan\theta = \frac{a_{13}}{a_{12}} = \frac{-1}{3}$$

$$\theta = -0.3218$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9487 & 0.3163 \\ 0 & -0.3163 & 0.9487 \end{bmatrix}$$

$$B = S_1^{-1}AS_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9487 & 0.3163 \\ 0 & -0.3163 & 0.9487 \end{bmatrix}^{-1} \begin{bmatrix} 2 & 3 & -1 \\ 3 & 1 & 2 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.9487 & 0.3163 \\ 0 & -0.3163 & 0.9487 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 3.1623 & 0 \\ 3.1623 & -0.4 & 2.2 \\ 0 & 2.2 & 0.4 \end{bmatrix}$$

$$|B - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 3.1623 & 0 \\ 3.1623 & -0.4-\lambda & 2.2 \\ 0 & 2.2 & 0.4-\lambda \end{vmatrix} = 0$$

The Sturm sequence is given by

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda = 2 - \lambda$$

$$P_2(\lambda) = (a_2 - \lambda)P_1(\lambda) - b_2^2 P_0(\lambda)$$

$$= (-0.4 - \lambda)(2 - \lambda) - (3.1623)^2(1)$$

$$P_2(\lambda) = \lambda^2 - 1.6\lambda - 10.8$$

$$P_3(\lambda) = (a_3 - \lambda)P_2(\lambda) - b_3^2 P_1(\lambda)$$

$$= (0.4 - \lambda)(\lambda^2 - 1.6\lambda - 10.8) - (2.2)^2(2 - \lambda)$$

$$\times (P_3(\lambda) = \lambda^3 - (2 - 0.4 + 0.4)\lambda^2 + (-5 - 3 -)\lambda$$

$$P_3(\lambda) = \lambda^3 - 2\lambda^2 - 15\lambda + 14 = 0 \quad -5 + 0.8 +$$

$$\lambda^3 - (\text{Sum of diag ele})\lambda^2 + (\text{Sum of minor diag. elements})\lambda^2 - \det(A) = 0$$

$$\lambda^3 - (2 - 0.4 + 0.4)\lambda^2 + \left(\begin{vmatrix} -0.4 & 2.2 \\ 2.2 & 0.4 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & 0.4 \end{vmatrix} + \begin{vmatrix} 3.1623 & 0.4 \\ 3.1623 & -0.4 \end{vmatrix} \right) = 0$$

$$- [2(-0.4 \times 0.4 - 2.2 \times 2.2) - 3.1623(3.1623 \times 0.4 - 0) + 0] = 0$$

$$\lambda^3 - 2\lambda^2 + (-15)\lambda - (-14) = 0$$

$$\lambda^3 - 2\lambda^2 - 15\lambda + 14 = 0$$

$$\lambda_1 = -3.4753$$

$$\lambda_2 = 4.5995$$

$$\lambda_3 = 0.8758$$

2. $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ to the diagonal form by given's method. Hence find the eigen value of A.

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\tan \theta = \frac{a_{13}}{a_{12}} = \frac{4}{2} = 2$$

$$\theta = 1.1071$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.4473 & -0.8944 \\ 0 & 0.8944 & 0.4473 \end{bmatrix}$$

$$B = S_1^{-1} A S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.4473 & -0.8944 \\ 0 & 0.8944 & 0.4473 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.4473 & -0.8944 \\ 0 & 0.8944 & 0.4473 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4.4722 & 0 \\ 4.4722 & 2.6002 & -1.997 \\ 0 & -1.997 & -0.6002 \end{bmatrix}$$

$$|B - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 4.4722 & 0 \\ 4.4722 & 2.6002-\lambda & -1.997 \\ 0 & -1.997 & -0.6002-\lambda \end{vmatrix} = 0$$

The Sturm sequence is given by -

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = a_1 - \lambda = 1 - \lambda$$

$$P_2(\lambda) = (a_2 - \lambda)P_1(\lambda) - b_2^2 P_0(\lambda)$$

$$= (2.6 - \lambda)(1 - \lambda) - (4.4722)^2(1)$$

$$P_2(\lambda) = 2.6 - 2.6\lambda - \lambda + \lambda^2 - 20.0006$$

$$P_2(\lambda) = \lambda^2 - 3.6\lambda - 17.4004$$

$$P_3(\lambda) = (a_3 - \lambda)P_2(\lambda) - b_3^2 P_1(\lambda)$$

$$= (-0.6 - \lambda)(\lambda^2 - 3.6\lambda - 17.4) - (-1.997)^2(1 - \lambda)$$

$$P_3(\lambda) = \lambda^3 - 3\lambda^2 + (-2.9994 - 0.6002 - 17.4004)\lambda - 9.0011$$

$$P_3(\lambda) = \lambda^3 - 3\lambda^2 - 21\lambda - 9.0011$$

$$P_3(\lambda) = \lambda^3 - 3\lambda^2 - 21\lambda - 9.0011 \neq 0$$

$$\lambda_1 = 6.4641$$

$$\lambda_2 = -0.4642$$

$$\lambda_3 = -3$$

2x2 - A6
3x3 - A4

* Rut's hauser method for arbitrary matrix:

$$A = A_1$$

$$A_1 = L_1 U_1$$

where $l_{ij} = 1$

$$L_1 = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix}$$

$$U_1 = \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$A_2 = U_1 L_1$$

$$A_2 = L_2 U_2$$

$$A_3 = U_2 L_2$$

$$A_3 = L_3 U_3$$

⋮

1. $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$

$$\text{Let } A_1 = L_1 U_1 = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$\therefore u_{11} = 4$$

$$u_{12} = 2$$

$$l_{21}u_{11} = -1$$

$$l_{21}u_{12} + u_{22} = 1$$

$$l_{21} = \frac{-1}{4}$$

$$\frac{-1}{4} \times 2 + u_{22} = 1 \Rightarrow u_{22} = \frac{3}{2}$$

$$A_2 = U_1 L_2$$

$$A_2 = \begin{bmatrix} 4 & 2 \\ 0 & 3/2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/4 & 1 \end{bmatrix} = \begin{bmatrix} 7/2 & 2 \\ -3/8 & 3/2 \end{bmatrix}$$

$$A_2 = L_2 U_2$$

$$\begin{bmatrix} 7/2 & 2 \\ -3/8 & 3/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{21} \end{bmatrix}$$

$$\begin{bmatrix} 7/2 & 2 \\ -3/8 & 3/2 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$\therefore U_{11} = 7/2$$

$$U_{12} = 2$$

$$L_{21}U_{11} = -3/8$$

$$L_{21}U_{12} + U_{22} = 3/2$$

$$L_{21} = \frac{-3 \times 2}{84 \times 7}$$

$$\frac{-3 \times 2}{28 \times 14} + U_{22} = \frac{3}{2}$$

$$L_{21} = \frac{-3}{28}$$

$$U_{22} = \frac{12}{7}$$

$$A_3 = U_2 L_2$$

$$A_3 = \begin{bmatrix} 7/2 & 2 \\ 0 & 12/7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3/28 & 1 \end{bmatrix} = \begin{bmatrix} 23/7 & 2 \\ -9/49 & 12/7 \end{bmatrix}$$

$$A_3 = L_3 U_3$$

$$A_3 = \begin{bmatrix} U_{11} & U_{12} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} \end{bmatrix}$$

$$\begin{bmatrix} 23/7 & 2 \\ -9/49 & 12/7 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} \end{bmatrix}$$

$$\therefore U_{11} = \frac{23}{7}$$

$$U_{12} = 2$$

$$L_{21}U_{11} = \frac{-9}{49}$$

$$L_{21}U_{12} + U_{22} = \frac{12}{7}$$

$$L_{21} = \frac{-9}{161}$$

$$U_{22} = \frac{42}{23}$$

$$A_4 = U_3 L_3$$

$$A_4 = \begin{bmatrix} 23/7 & 2 \\ 0 & 42/23 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -9/161 & 1 \end{bmatrix} = \begin{bmatrix} 73/23 & 2 \\ -54/529 & 42/23 \end{bmatrix}$$

$$A_4 = L_4 U_4$$

$$A_4 = \begin{bmatrix} U_{11} & U_{12} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} \end{bmatrix}$$

$$\begin{bmatrix} 73/23 & 2 \\ -54/529 & 42/23 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} \\ L_{21}U_{11} & L_{21}U_{12} + U_{22} \end{bmatrix}$$

$$\therefore U_{11} = \frac{73}{23}$$

$$U_{12} = 2$$

$$l_{21}u_{11} = \frac{-54}{529}$$

$$l_{21}u_{12} + u_{22} = \frac{42}{23}$$

$$l_{21} = \frac{-54}{1679}$$

$$u_{22} = \frac{138}{73}$$

$$A_5 = U_4 L_4$$

$$A_5 = \begin{bmatrix} 73/23 & 2 \\ 0 & 138/73 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -54/1679 & 1 \end{bmatrix} = \begin{bmatrix} 227/73 & 2 \\ -324/5329 & 138/73 \end{bmatrix}$$

$$A_5 = \begin{bmatrix} 3.1095 & 2 \\ -0.06 & 1.8904 \end{bmatrix}$$

$$A_5 = L_5 U_5$$

$$A_5 = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$\begin{bmatrix} 3.1095 & 2 \\ -0.06 & 1.8904 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} \end{bmatrix}$$

$$\therefore u_{11} = 3.1095$$

$$u_{12} = 2$$

$$l_{21}u_{11} = -0.06$$

$$l_{21}u_{12} + u_{22} = 1.8904$$

$$l_{21}u_{11} = -0.0193$$

$$u_{22} = 1.9289$$

$$A_6 = U_5 L_5$$

$$A_6 = \begin{bmatrix} 3.1095 & 2 \\ 0 & 1.9289 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -0.0193 & 1 \end{bmatrix} = \begin{bmatrix} 3.0709 & 2 \\ -0.037 & 1.9289 \end{bmatrix}$$

2. Find all the eigen values of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$. Using Rutis hauser method iterate till the lower triangular parts are less than 0.05 in magnitude.

$$A_1 = L_1 U_1$$

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$\therefore u_{11} = 1, \quad u_{12} = 1, \quad u_{13} = 1$$

$$l_{21}u_{11} = 2 \quad l_{21}u_{12} + u_{22} = 1 \quad l_{21}u_{13} + u_{23} = 2$$

$$l_{21} = 2 \quad 2 + u_{22} = 1 \quad 2 + u_{23} = 2$$

$$u_{22} = -1 \quad u_{23} = 0$$

$$l_{31}u_{11} = 1 \quad l_{31}u_{12} + l_{32}u_{22} = 3 \quad l_{31}u_{13} + l_{32}u_{23} + u_{33} = 2$$

$$l_{31} = 1 \quad 1 - l_{32} = 3 \quad 1 + 0 + u_{33} = 2$$

$$l_{32} = -2 \quad u_{33} = 1$$

$$A_2 = U_1 L_1$$

$$A_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -1 & 1 \\ -2 & -1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

$$A_2 = L_2 U_2$$

$$\begin{bmatrix} 4 & -1 & 1 \\ -2 & -1 & 0 \\ 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$\therefore u_{11} = 4, \quad u_{12} = -1, \quad u_{13} = 1$$

$$l_{21}u_{11} = -2 \quad l_{21}u_{12} + u_{22} = -1 \quad l_{21}u_{13} + u_{23} = 0$$

$$l_{21} = -1/2 \quad 0.5 + u_{22} = -1 \quad -0.5 + u_{23} = 0$$

$$l_{21} = -0.5 \quad u_{22} = -1.5 \quad u_{23} = 0.5$$

$$l_{31}u_{11} = 1 \quad l_{31}u_{12} + l_{32}u_{22} = -2 \quad l_{31}u_{13} + l_{32}u_{23} + u_{33} = 1$$

$$l_{31} = 0.25 \quad -0.25 - l_{32} \times 1.5 = -2 \quad u_{33} = 0.17$$

$$l_{32} = 1.17$$

$$A_3 = U_2 L_2$$

$$A_3 = \begin{bmatrix} 4 & -1 & 1 \\ 0 & -1.5 & 0.5 \\ 0 & 0 & 0.17 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -0.5 & 1 & 0 \\ 0.25 & 1.17 & 1 \end{bmatrix} = \begin{bmatrix} 4.75 & 0.17 & 1 \\ 0.88 & -0.92 & 0.5 \\ 0.04 & 0.2 & 0.17 \end{bmatrix}$$

$$A_3 = L_3 U_3$$

$$\begin{bmatrix} 4.75 & 0.17 & 1 \\ 0.88 & -0.92 & 0.5 \\ 0.04 & 0.2 & 0.17 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ l_{21}U_{11} & l_{21}U_{12} + U_{22} & l_{21}U_{13} + U_{23} \\ l_{31}U_{11} & l_{31}U_{12} + l_{32}U_{22} & l_{31}U_{13} + l_{32}U_{23} + U_{33} \end{bmatrix}$$

$$U_{11} = 4.75$$

$$U_{12} = 0.17$$

$$U_{13} = 1$$

$$l_{21}U_{11} = 0.88$$

$$l_{21}U_{12} + U_{22} = -0.92$$

$$l_{21}U_{13} + U_{23} = 0.5$$

$$l_{21} = 0.19$$

$$U_{22} = -0.95$$

$$U_{23} = 0.31$$

$$l_{31}U_{11} = 0.04$$

$$l_{31}U_{12} + l_{32}U_{22} = 0.2$$

$$l_{31}U_{13} + l_{32}U_{23} + U_{33} = 0$$

$$l_{31} = 0.01$$

$$l_{32} = -0.21$$

$$U_{33} = 0.23$$

$$A_4 = U_3 L_3$$

$$A_4 = \begin{bmatrix} 4.75 & 0.17 & 1 \\ 0 & -0.95 & 0.31 \\ 0 & 0 & 0.23 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0.19 & 1 & 0 \\ 0.01 & -0.21 & 1 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} 4.79 & -0.04 & 1 \\ -0.18 & -1.02 & 0.31 \\ 0 & -0.05 & 0.23 \end{bmatrix}$$

$$A_4 = L_4 U_4$$

$$\begin{bmatrix} 4.79 & -0.04 & 1 \\ -0.18 & -1.02 & 0.31 \\ 0 & -0.05 & 0.23 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ l_{21}U_{11} & l_{21}U_{12} + U_{22} & l_{21}U_{13} + U_{23} \\ l_{31}U_{11} & l_{31}U_{12} + l_{32}U_{22} & l_{31}U_{13} + l_{32}U_{23} + U_{33} \end{bmatrix}$$

STILL CONTINUES

* Householder's method :

$$A_1 = []_{4 \times 4}$$

Ist notation :

$$A_2 = S_1 A_1 S_1$$

$$S_1 = I - 2V_1 V_1^T$$

$$V_1 = [0 \ S_2 \ S_3 \ S_4]^T \rightarrow [0 \ S_2 \ S_3]^T$$

$$S_2^2 = \frac{1}{2} \left[1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times S_2}$$

$$S_4 = \frac{a_{14} \times \text{sign}(a_{12})}{2q \times S_2} \rightarrow \text{eliminate}$$

$$q = \sqrt{a_{12}^2 + a_{13}^2 + a_{14}^2} \rightarrow q = \sqrt{a_{12}^2 + a_{13}^2}$$

IInd notation :

$$A_3 = S_2 A_2 S_2$$

$$S_2 = I - 2V_2 V_2^T$$

$$V_2 = [0 \ 0 \ S_3 \ S_4]^T$$

$$S_3^2 = \frac{1}{2} \left[1 + \frac{a_{23} \times \text{sign}(a_{23})}{q} \right]$$

$$S_4 = \frac{a_{24} \times \text{sign}(a_{23})}{2q \times S_3}$$

$$q = \sqrt{a_{23}^2 + a_{24}^2}$$

1. Use the householder method to reduce the matrix $A = \begin{bmatrix} 2 & -1 & -1 & 1 \\ -1 & 4 & 1 & -1 \\ -1 & 1 & 3 & -1 \\ 1 & -1 & -1 & 2 \end{bmatrix}$ into the tridiagonal form.

$$A_1 = A$$

Ist notation :

$$A_2 = S_1 A_1 S_1$$

$$S_1 = I - 2V_1 V_1^T$$

$$V_1 = [0 \ S_2 \ S_3 \ S_4]^T$$

$$q = \sqrt{a_{12}^2 + a_{13}^2 + a_{14}^2}$$

$$q = \sqrt{(-1)^2 + (-1)^2 + 1^2} = \sqrt{3}$$

$$S_2^2 = \frac{1}{2} \left[1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$S_2^2 = \frac{1}{2} \left[1 + \frac{(-1) \times (-)}{\sqrt{3}} \right]$$

$$S_2 = 0.8881$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times S_2} = \frac{(-1) \times (-)}{2(\sqrt{3}) \times 0.8881}$$

$$S_3 = 0.3250$$

$$S_4 = \frac{a_{14} \times \text{sign}(a_{12})}{2q \times S_2} = \frac{1 \times (-)}{2\sqrt{3} \times 0.8881}$$

$$S_4 = -0.3250$$

$$V_1 = [0 \quad 0.8881 \quad 0.3250 \quad -0.3250]^T$$

$$S_1 = I - 2V_1V_1^T$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 0.8881 \\ 0.3250 \\ -0.3250 \end{bmatrix} \begin{bmatrix} 0 & 0.8881 & 0.3250 & -0.3250 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0.7887 & 0.2886 & -0.2886 \\ 0 & 0.2886 & 0.1056 & -0.1056 \\ 0 & -0.2886 & -0.1056 & 0.1056 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.5774 & -0.5774 & 0.5774 \\ 0 & -0.5774 & 0.7887 & 0.2113 \\ 0 & 0.5774 & 0.2113 & 0.7887 \end{bmatrix}$$

$$A_2 = S_1 A_1 S_1$$

$$A_2 = \begin{bmatrix} 2 & 1.732 & 0 & 0 \\ 1.732 & 5 & 0.2113 & -0.7787 \\ 0 & 0.2113 & 2.8867 & -0.5 \\ 0 & -0.7787 & -0.5 & 1.7113 \end{bmatrix}$$

II rotation :

$$A_3 = S_2 A_2 S_2$$

$$S_2 = I - 2V_2 V_2^T$$

$$V_2 = [0 \ 0 \ s_3 \ s_4]^T$$

$$s_3^2 = \frac{1}{2} \left[1 + \frac{a_{23} \times \text{sign}(a_{23})}{q} \right]$$

$$q = \sqrt{a_{23}^2 + a_{24}^2} = \sqrt{(0.2113)^2 + (-0.7787)^2}$$

$$q = 0.8069$$

$$s_3^2 = \frac{1}{2} \left[1 + \frac{(0.2113) \times (+)}{0.8069} \right]$$

$$s_3 = 0.7943$$

$$s_4 = \frac{a_{24} \times \text{sign}(a_{23})}{2q \times s_3} = \frac{-(0.7787) \times (+)}{2(0.8069) \times 0.7943}$$

$$s_4 = -0.6076$$

$$V_2 = [0 \ 0 \ 0.7943 \ -0.6076]^T$$

$$S_2 = I - 2V_2 V_2^T$$

$$S_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 0 \\ 0.7943 \\ -0.6076 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0.7943 & -0.6076 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -0.2618 & 0.9652 \\ 0 & 0 & 0.9652 & 0.2618 \end{bmatrix}$$

$$A_3 = S_2 A$$

$$A_3 =$$

2. Using Householder method, reduce the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ into tridiagonal form and hence find the eigen values.

$$A_1 = A$$

I notation :

$$A_2 = S_1 A_1 S_1$$

$$S_1 = I - 2V_1 V_1^T$$

$$V_1 = [0 \ S_2 \ S_3]^T$$

$$S_2^2 = \frac{1}{2} \left[1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$q = \sqrt{a_{12}^2 + a_{13}^2} = \sqrt{1+1} = \sqrt{2}.$$

$$S_2^2 = \frac{1}{2} \left[1 + \frac{1 \times (+)}{\sqrt{2}} \right]$$

$$S_2 = 0.9239.$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times S_2}$$

$$S_3 = \frac{1 \times (+)}{2\sqrt{2} \times 0.9239} = 0.3827.$$

$$V_1 = [0 \ 0.9239 \ 0.3827]^T$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 0.9239 \\ 0.3827 \end{bmatrix} \begin{bmatrix} 0 & 0.9239 & 0.3827 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.7072 & -0.7072 \\ 0 & -0.7072 & 0.7072 \end{bmatrix}$$

$$A_2 = S_1 A_1 S_1$$

$$A_2 = \begin{bmatrix} 2 & -1.4144 & 0 \\ -1.4144 & 1.0002 & 0 \\ 0 & 0 & 1.0002 \end{bmatrix}$$

$$|A_2 - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -1.4144 & 0 \\ -1.4144 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

The sturm sequence is given by

$$P_0(\lambda) = 1$$

$$P_1(\lambda) = 2 - \lambda$$

$$\begin{aligned} P_2(\lambda) &= (a_2 - \lambda) P_1(\lambda) - b_2^2 P_0(\lambda) \\ &= (1 - \lambda)(2 - \lambda) - (-1.41)^2(1) \\ &= 2 - 2\lambda - \lambda + \lambda^2 - 1.9881 \\ &= \lambda^2 - 3\lambda + 0.0119 \end{aligned}$$

$$P_3(\lambda) = \lambda^3 - 4\lambda^2 + 3\lambda - 0$$

$$\lambda_1 = 3 \quad \& \quad \lambda_2 = 1, \quad \lambda_3 = 0$$

3.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$$

$$A_1 = A$$

$$A_2 = S_1 A_1 S_1$$

$$S_1 = I - 2v_1 v_1^T$$

$$v_1 = [0 \quad s_2 \quad s_3]^T$$

$$q = \sqrt{a_{12}^2 + a_{13}^2}$$

$$q = \sqrt{4 + 9} = \sqrt{13}$$

$$S_2^2 = \frac{1}{2} \left[1 + \frac{a_{12} \times \text{sign}(a_{12})}{q} \right]$$

$$S_2^2 = \frac{1}{2} \left[1 + \frac{2 \times (+)}{\sqrt{3}} \right]$$

$$S_2 = 0.8817$$

$$S_3 = \frac{a_{13} \times \text{sign}(a_{12})}{2q \times S_2}$$

$$S_3 = \frac{3 \times (+)}{2 \times \sqrt{3} \times 0.8817} = 0.4718$$

$$V_1 = [0 \quad 0.8817 \quad 0.4718]^T$$

$$S_1 = I - 2V_1V_1^T$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 0.8817 \\ 0.4718 \end{bmatrix} \begin{bmatrix} 0 & 0.8817 & 0.4718 \end{bmatrix}$$

$$S_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.5548 & -0.832 \\ 0 & -0.8320 & 0.5548 \end{bmatrix}$$

$$A_2 = S_1 A_1 S_1$$

$$A_2 = \begin{bmatrix} 1 & -3.6055 & 0 \\ -3.6055 & 0.0768 & -0.3844 \\ 0 & -0.3844 & 1.9232 \end{bmatrix}$$

Eigen values:

$$\lambda^3 - 3\lambda^2 - 11\lambda + 25 = 0$$

$$\lambda_1 = -3.11, \quad \lambda_2 = 4.20, \quad \lambda_3 = 1.91$$